

VISHAL KUMAR

✉ vishalku@usc.edu  linkedin.com/in/vishal-kumar-130a90249  github.com/vishxlshxrma  yt-portfolio-xi.vercel.app

Education

University of Southern California

Master's of Science in Computer Science

2025 – 2026

Los Angeles, CA

Guru Gobind Singh Indraprastha University

B.Tech. in Computer Science

2020 – 2024

New Delhi, India

Experience

Teaching Assistant, USC Viterbi K-12 STEM Center

Summer 2025 & Summer 2026

University of Southern California

Los Angeles, CA

- Delivered **200+** hours of instruction in Drones, Python, and Game Development to **300+** high school students as part of USC's K-12 outreach.
- Designed structured learning modules and projects, increasing student task completion rates by **25%**.
- Guided students during labs and office hours, helping debug code and reinforce core CS concepts and programming fundamentals.

Software Developer Intern

Jun 2024 – Sep 2024

Punjab National Bank

New Delhi, India

- Developed and deployed a Litigation Management System using Java Servlets, JSP, and MySQL, streamlining case tracking and document workflows.
- Built RESTful APIs using Node.js and Express to automate banking processes, reducing processing time by **40%**.
- Optimized backend logic and database queries, improving response time by **50%** under **~500** concurrent users and reducing errors by **25%**.
- Deployed on Apache Tomcat and collaborated in Agile sprints, improving release efficiency by **15%**.

Research Publications

Generative A.I. Integration using APIs

2023

- Published in Satyam, official journal of GGSIPU, under guidance of Dr. Sangeeta and Dr. Sitender.
- Conducted literature-driven research on Large Language Model (LLM) and API integration strategies, focusing on reliability, scalability, and practical software patterns.
- Synthesized best practices for designing AI-augmented workflows without introducing new model architectures, emphasizing real-world applicability.

Technical Skills

Languages: Python, JavaScript, Java, Go, Swift, SQL/MySQL, C#

ML & Data: TensorFlow, YOLOv8, NumPy, pandas, scikit-learn

Frameworks: Express, Angular, FastAPI, Node.js, React, Next.js, Celery, Tailwind CSS

Databases: PostgreSQL, MongoDB, Redis, Firebase, MySQL

Tools: Docker, RabbitMQ, Celery, REST APIs, Git/GitHub, Copilot, Hibernate, Base44, Huggingface

Projects

Transizr — AI-Powered Media Transcription Platform | *FastAPI, Celery, RabbitMQ, Redis, Docker*

2026

- Developed a distributed transcription system using containerized workers and message queues for scalable audio/video-to-text processing.
- Built Celery-RabbitMQ pipelines enabling parallel, fault-tolerant task execution across multiple containers.
- Added real-time job tracking using Redis Pub-Sub and Server-Sent Events (SSE) for instant UI updates.

Artsy Platform - Web App | *Angular, Node.js, mongoDB, JWT Auth*

2025

- Developed a scalable full-stack web application using Angular, TypeScript, Node.js, Express.js, and REST APIs for real-time artist and artwork discovery.
- Integrated the Artsy API with asynchronous data fetching, dynamic search, artwork insights, and similarity-based recommendations.
- Implemented JWT authentication, MongoDB, state management, responsive UI, and component-based architecture for secure personalized favorites.

CodeBundle — Directory-to-Text Export CLI | *Go, CLI*

2026

- Built a Go-based CLI that bundles entire codebases into a structured .txt file for LLM workflows and code review.
- Added a high-speed *Fast Mode* that optimizes file collection for large repositories.